

40 — Findings Register, Risks & Task List

v1.6 — the honest state of the program, both tracks

Function: The consolidated register of every finding that materially changed the numbers or the architecture — solid-track F1–F6 and cross-track X1–X12, X14 — plus the correction trail, the task list, and the make-or-break bench questions. **Anyone reconstructing this concept should read this document before trusting any sizing figure.** The physics backbone holds on multiple independent passes; quantitative closure is bench-gated exactly where marked.

Terminology. *Dew-point indirect evaporative stage / dew-point IEC* is the same stage as the *Maisotsenko cycle / M-cycle* used in the finding statements below (doc 00 §3). The finding statements are unchanged.

1. Solid-track findings (F-series)

F1 — M-cycle working-air moisture is the dominant latent term

The sink (29 °C) cannot absorb heat from a 25 °C cabin, so all cabin sensible load exits evaporatively via the M-cycle working air; that moisture must pass through the desiccant (exhaust-and-replace in open cycle, recycle in X8 mode — equivalent within ~10–16% at DP-A). **Effect: peak duty ~9–11 kg/h, not ~2 kg/h; regen ~7.5–9 kW continuous; condenser/fan duty scale with it.** The self-consistent steady state (doc 00 §4) sharpened the original 7–8 kg/h estimate — the hand-chained cabin state was not self-consistent. Closure: the moisture returns as condensate covering M-cycle feed with ~50–70 L/day surplus; the system is a coherent heat-driven chiller. Full transient T2 sets the final charge.

F2 — Solar-direct regeneration is equilibrium-dead at the peak point

Against a condensing purge (~25 g/kg), a 45–50 °C bed faces ~29–38% RH at its face — at/above AlFu's step: zero driving force; no kinetic result rescues it (table, doc 20 §6). Heat-source decision: waste-heat primary; the **liquid track is the solar comfort island** (X2), with heat-pump lift retained behind it. M2 *confirms* F2; its real deliverables are 60–65 °C completeness/kinetics and effective Δq .

F3 — The sizing Δq is optimistic

0.2–0.3 g/g is a **dry-purge full-swing** figure; a plausible 0.05–0.10 g/g residual at 60–65 °C against humid purge cuts effective Δq to ~0.15–0.2 g/g (+25–50% inventory), compounding F1. All sizing tables carry the annotation; M2 extracts effective Δq as a first-class output.

F4 — Mill-zero needs a feedstock-reactivity branch

The literature mortar-and-pestle demonstration used *freshly precipitated* amorphous $\text{Al}(\text{OH})_3$; crystalline tech-grade gibbsite is kinetically sluggish. A PXRD fail may be feedstock, not milling energy. Mitigations, cheapest first: sieve fine → extend aging toward 120 °C → **dissolve-and-reprecipitate fresh gel** (an evening vs a mill build). Integrated into the doc 21 §4 tree.

F5 — Aluminum DCHX in chloride service

The coated aluminum exchanger sits in salt-bearing intake air with brine and raw water elsewhere on the platform — the materials class the two-worlds rule prohibits near chloride. Mitigations decided on paper (doc 22 §3): sealed/filtered intake, no dissimilar-metal fittings, coating treated as a barrier layer with edge/defect behavior in M4's scope, materials law imported from the liquid track — and **X8 closure as the primary mitigation** (the working path ingests no ambient aerosol). Corrosion *rate* is M4's.

F6 — The DCHX sensible-cycling bucket is unverified

The sensible bed-cycling allowance in doc 20 §3 is a sizing-grade assumption, not a measurement. It is bounded by the total swung heat capacity — coated-plate metal, binder, sorbent, and the fluid inventory resident in the module at swap — divided by the water swung per half-cycle. No plate thickness, channel pitch, or per-module fluid inventory is specified anywhere in the lineage, so the ratio cannot be evaluated and **the magnitude is OPEN**.

If the allowance proves optimistic, the specific-energy line and the continuous regeneration line in doc 20 §3/§5 both move upward, and bed-to-bed heat recovery moves from a design response to a hard requirement. Gated on **M3** (blank run, geometry fixed) and on **M1's** resolution of the coated-area denominator, which changes sorbent-per-unit-metal depending on one-side vs two-side coating and so gates this entry as well as the area bookkeeping it is currently filed under. General mechanism: **X14. No headline figure is altered by this entry.**

Minor notes

- **PVA anneal:** the 120–150 °C activation doubles as the mandatory anneal (doc 22 §2) — a re-coat never skips it.
- **Ten-minute cycling** thermally cycles the fluid inventory and manifolds; the sensible bucket may be optimistic for the plumbing (doc 20 §4; M3 blank measures it) — now recorded as **F6**, with the general mechanism at **X14**.

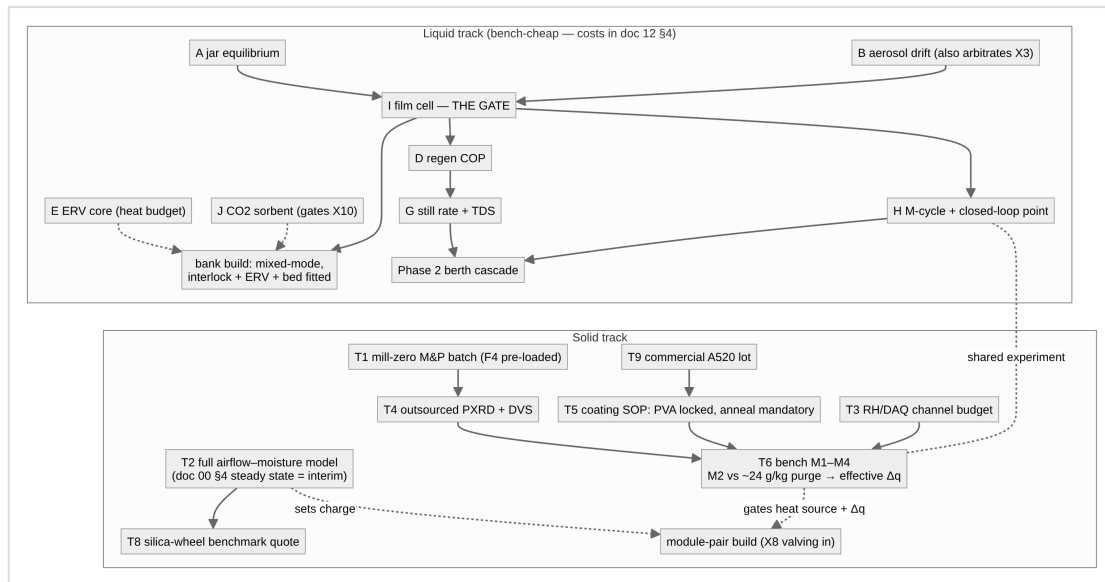
2. Cross-track findings (X-series)

#	Finding (one line)	Where it lives now
X1	Once-through ventilation and whole-cabin M-cycle cooling never compose (5.5–11 kg/h absorber duty at DP-A); whole-cabin liquid AC deferred — in recirc topology it converges to solid-track-class duty (~10.6 kg/h, ~11 kW)	doc 00 §4/§6; doc 10 §2
X2	Regeneration asymmetry is a theorem: the sealed still (no purge) keeps positive driving force at any pool >~40 °C; the solid bed hits F2's wall below ~50 °C → liquid = solar layer, solid = waste-heat layer	doc 00 §3; doc 30 §2
X3	The old blanket rejection of liquid desiccants for cabin air is re-scoped: "rejected absent demonstrated aerosol control — test B is the arbiter"	doc 21 §2; doc 11 §3
X4	Materials doctrine flows both ways: nylon ban, brazed-plate prohibition, drip discipline, sourcing heuristic imported to the solid track	doc 22 §3
X5	The airflow cascade: route makeup through the cabin before it becomes working air — ventilation becomes latently free. Berth scale: full 48 m³/h kept (~1,920 vs ~2,670 ppm), <i>more</i> cooling (102 vs 86 W), 1.6 °C supply cost. Solid open cycle: ~600 ppm automatic at F1-scale flows	doc 10 §2; doc 20 §2
X6	Mixed-mode recirculation (CO ₂ -interlocked fresh minimum + recirculated bulk) is the liquid occupied baseline — once-through fails 4-adult comfort at DP-A (59–66% RH); mixed-mode restores 55% at 0.88 kg/h. Recirc-only prohibited	doc 10 §2; doc 00 §5
X7	ERV latent recovery on the fresh stream: $(1-\epsilon) \times 10.4 \text{ g/kg per kg}$; $\epsilon \geq 0.8$ saves ~9–10 kWh/day at DP-A and makes generous ventilation cheap. Requires one ducted exhaust path; CO ₂ crossover <5% (test E). Applies to both tracks	doc 11 §5; doc 20 §2
X8	Exhaust-vapor recovery doctrine: saturated process exhausts terminate in distillation recovery; no stream is both rejection sink and recovery source (protects the ERV); closed working-air loop = solid design intent (water-neutral in every ambient) and liquid free-heat option (~1.3 kWh/L); air-swept regen gains a condenser (technical water PENDING TDS)	doc 00 §7; doc 20 §8; doc 11 §4
X9	All-electric galley, no gas in the envelope: one burner = 3.6× the crew's CO ₂ + ~0.25 kg/h latent. ~2 kWh _e /day induction	doc 00 §5; doc 30 §4
X10	The CO ₂ battery: two-bed solid-amine TSA, 85–95 °C regen, ~1.6 kg/day for ~1.6–2.5 kWh/day — the only path holding <1,000 ppm sealed. Required-PENDING test J (slip assay gates cabin connection); fallback = oversized ERV+DCV, open-cycle only	doc 00 §5; doc 11 §5; doc 12 §4
X11	Sealed-mode CO ₂ -bed chemistry is heat-grade-dependent: K ₂ CO ₃ /apolar-carbon TSA (regen ~130–150 °C; alumina/MgO supports prohibited — double-salt deactivation; breathing-air gate = alkaline-carryover assay, required-PENDING test J-K) is the waste-heat-platform primary; the solid-amine resin (85–95 °C) is the solar-grade bed. The ladder runs ventilation-first from the top of heat availability down; a small amine bed is retained as outage fallback where potash is primary (the ETC's ≤93 °C cannot regenerate carbonate)	doc 00 §5; doc 12 §4–6; doc 30 §2/§6
X12	The brine's activity depression is a temperature lift , not only a humidity floor (the dual of X2): a single-stage CaCl ₂ absorption heat transformer — with the sealed still unchanged as its generator + condenser — upgrades a 60–65 °C waste tail to 85–90 °C at COP ~0.45–0.48 with no electricity beyond transfer pumps, reaching amine-bed and hot-regen grade. Lift ceiling $aw(x) \cdot Psat(T_{abs}) < Psat(T_{evap}) \rightarrow \leq 91 \text{ °C}$ single-stage (potash grade out of reach). Upgrade path only, never load-bearing; PENDING test L	doc 31 §2; doc 12 §4
X14	Thermal-swing sensible penalty is set by the inert-mass ratio, not by cycle time. For any thermal-swing bed on the platform — the solid-track DCHX, the amine CO ₂ bed (X10), the potash bed (X11) — the sensible heat spent per unit of sorbate swung is the total swung heat capacity times the swing ΔT, divided by sorbent mass times working capacity. Half-cycle duration cancels: shortening the cycle raises throughput per unit of coated area without changing the specific penalty. The design levers are therefore the inert-mass-to-sorbent ratio (plate metal, binder, resident fluid) and bed-to-bed recovery effectiveness, never cycle speed . Magnitudes OPEN pending geometry (F6; M3)	doc 20 §3–4; doc 22 §1

X13 is reserved for the solubility-swing class-ceiling finding, drafted and pending. The gap is deliberate: X-numbers are assigned in the order findings enter the queue, and that order is part of the register's audit trail.

Specification P17 (safety-critical): CO₂ <1,000 ppm at all times, all modes; alarm + forced boost 2,000 ppm; per-room max sensing; mechanical minimum stop. Full register: doc 00 §8.

3. Dependency structure and task list



Ranking: liquid A/B/E/J and solid T1/T9 all run in parallel at trivial cost; test I is the liquid gate; T2 gates every solid downstream size. Shared: test H and the solid M-cycle validation are one experiment; the wet/dry-bulb instrumentation stack (doc 22 §7) serves both. **T7 (documentation pass) is closed by this lineage.** Deferred: CAU-23/CAU-10-H, the CO₂-sorbent watch register (doc 12 §6 — MOF-808-AA primary; tetraamine-Mg₂(dobpdc), superseding the diamine class), two-core sensible recovery, the hybrid brine-pre-stage-before-AIFu architecture (worth revisiting only with hot-regen brine — the polish split favors it only then). On waste-heat platform variants, test J-K joins the parallel cheap set (X11), sharing J's rig and instrumentation. The **doc 31 upgrade paths** (X12 AHT first; coupled VC heat pump; still MVR; closed AIFu chiller; static crystallizer pot) are boost modes on the deferred list — additions riding on the passive baseline, never core; tests **L** and **A3** (doc 12 §4) join the cheap parallel set only when their platform trigger exists (waste-heat tail / mass-limited reserve respectively), and the prime-mover heat-pump variant is recorded rejected (doc 31 §1).

Tooling task — the register-freshness gate does not gate what doc 50 §3.5 specifies. `scripts/check_release.py`'s `check_register()` compares modification times only: it fails when any `docs/*.md` is newer than `docs/parameter_register.xlsx`, and passes otherwise. It never compares the register's *contents* against the documents. A bare regeneration therefore turns it green whether or not the register still reproduces what the documents say. Doc 50 §3.5/§8 makes release-time reconciliation of the register's `Checks` sheet a publication gate, so the gate as implemented enforces staleness, not agreement — the reconciliation itself remains a human judgement item under doc 50 §9. Fixing it is its own change, not part of this lineage pass.

Tooling task (closed in v1.4) — the release gate could not see a broken render. Until v1.4 `scripts/check_release.py` verified that `rendered/` was complete, newer than its sources, and free of raw Mermaid — but never that a PDF paginated or carried page numbers. On 2026-08-27 an unpatched-Qt `wkhtmltopdf` rendered all twelve documents as single pages with everything past page 1 clipped off-canvas, and the whole set passed. It was caught by eye, not by the gate, and would otherwise have gone into the Zenodo deposit. Closed by `check_pdf_pagination()` plus an engine pin that makes `render_docs.sh` refuse to run unpinning (doc 50 §3.3). Recorded here because the near-miss is part of the correction trail: the gate's coverage is itself a thing that needs auditing.

4. Make-or-break bench questions (consolidated)

- ☐ Real CaCl₂ aw vs wt% and T — 40 wt% @ 33 °C ≤ 55% ERH (A)
- ☐ Aerosol chain: clean steel coupon, trickle and flooded (B — gates every cabin connection and arbitrates X3)
- ☐ Film K-a, wetting threshold, staged NTU/m, mixed-mode flow point (I)
- ☐ ERV real ϵ_{lat} , salt fouling, CO₂ crossover <5% (E)
- ☐ CO₂ sorbent: capacity at 1,000–1,500 ppm / 45–55% RH, 85–95 °C completeness, water co-adsorption, **amine/ammonia slip**, cycle fade (J)
- ☐ Potash arm (X11): K₂CO₃/carbon capacity at 1,000–1,500 ppm / 45–55% RH; 120/135/150 °C completeness with logged purge; alumina-deactivation control; caking/deliquescence; **alkaline carryover**; cycle stability (J-K)
- ☐ Effective working Δq under ~24 g/kg purge at 60–65 °C (F3 · M2)
- ☐ Desorption completeness/rate in the 10-min half-cycle (F2 · M2)
- ☐ Specific regeneration energy vs 0.8–1.0 kWh/L, incl. plumbing overhead (M3)
- ☐

- Coating adhesion, anneal, capacity retention, **coated-face ΔP** (M1, T5)
- ☐ Coating durability + salt-air edge behavior over 10^2 – 10^3 swings (M4 · F5)
- ☐ Still rate + condensate TDS ≈ 0 ; air-swept-condenser TDS (G)
- ☐ M-cycle wetting/heel + closed-loop feed point (H — shared)
- ☐ Distillate potability lab test before regular drinking
- ☐ T2 outputs: charge, coated area, condenser duty, fan power at the self-consistent peak; correction path (cycling vs area vs setpoint)
- ☐ **Coated-area denominator** — doc 22 §4's sizing table implies ~ 0.3 kg/m² where §2 specifies 0.18 kg/m² planform (one-side vs two-side convention unstated). Settled empirically by M1 on a known geometry; size from 0.18 until then (doc 22 §4)
- ☐ **CO₂-battery duty basis** — the ~ 1.6 kg/day figure is a 16 h/8 h daily average; the all-awake peak at the 48 m³/h floor is ~ 2.05 kg/day, and the published 91 m³/h at 50% capture independently implies ~ 2.16 kg/day. P17 is an at-all-times spec, so bed mass and regen heat want the peak (test J)
- ☐ **One envelope-leakage assumption for both tracks** — liquid sizes on 280 g/h (occupants only), solid on 1.8–2.8 kg/h (envelope + occupant), implying ~ 0.95 ACH where doc 12 §3 uses 0.05–0.15 for the unattended case. Bounded: duty moves 8.8→10.3 kg/h across that range, so F1 holds either way
- ☐ **Contactator rating stated with its approach and flow** — the 0.4–0.8 kg/h rating reproduces at the ~ 0.19 m/s DP-A baseline, not at the stated 0.6–1.0 m/s band; the NTU-1.9/85% approach used for depth does not hold at the higher velocity (test I)
- ☐ **Real ϵ_{lat} behind the ERV'd duty line** — doc 10 §3 and doc 12 §1 carry 17.4 and 15.9 g/kg for the same pre-dried state, i.e. $\epsilon \approx 0.65$ vs the specified ≥ 0.8 ; the published ERV'd removal tracks 17.4. Settled by test E (doc 12 §2 erratum 10). The bare-fresh 0.88 kg/h peak governs sizing either way

5. Correction-trail principle

Every erratum stays visible (doc 12 §2 carries the liquid trail; F1's sharpening from 7–8 to 9–11 kg/h is recorded above). The correction trail is part of the design's credibility: *never cite a margin as a computed value; re-run every comfort claim when the design point moves; solve steady states simultaneously, never as hand chains.*

The parameter register (doc 50 §3.5) is the standing mechanism for this: it re-derives every headline figure from its stated basis, so a number that has drifted from the basis printed beside it surfaces on its Checks sheet rather than surviving unexamined. Reconciling that sheet is a release gate (doc 50 §8). Its first pass produced the two open items above, erratum 10, and a set of basis-labelling clarifications across docs 00/10/12/20/22 — no design figure moved. *Added lesson: a quantity is only as good as the basis printed next to it; state the basis, or the number will eventually be read on the wrong one.*

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Version history

- **v1.0** — New lineage: archived 05 (F1–F4) merged with the archived doc 06 X-register (X1–X10, F5, P-series patches now applied in place across docs 00–30); task T7 closed by this lineage; dependency chart unified across tracks.
- **v1.1** — X11 recorded; test J-K added to the make-or-break list and task notes; deferred list updated to the doc 12 §6 watch register.
- **v1.2** — X12 (CaCl₂ absorption heat transformer) recorded; doc 31 upgrade-path family added to the deferred notes with tests L/A3 pointers and the prime-mover heat-pump rejection; X-range extended in the function statement.
- **v1.3** — First parameter-register reconciliation pass (doc 50 §3.5): two open items added to the make-or-break list — the doc 22 coated-area denominator (gated on M1) and the ϵ_{lat} inconsistency behind the ERV'd duty line (gated on test E, doc 12 §2 erratum 10) — and §5 records the register as the standing mechanism for catching basis drift. No finding ID renumbered; no design figure changed.
- **v1.4** — Dependency-chart label follows doc 12 v1.4 in dropping the retired aggregate cost headline; four verification findings added to §4 (CO₂-battery duty basis, the X2 crossover, the cross-track latent-gains basis, and the contactator rating/velocity/approach triangle).
- **v1.5** — **F6** recorded (the DCHX sensible-cycling bucket is unverified; magnitude OPEN, gated on M3 and on M1's coated-area denominator) and **X14** recorded (thermal-swing sensible penalty is set by the inert-mass ratio, not by cycle time — half-cycle duration cancels). **X13 is reserved** for the solubility-swing class-ceiling finding, drafted and pending; the gap is deliberate and is part of the queue's audit trail. Dew-point IEC / M-cycle equivalence pointer added; F/X ranges extended in the function statement. §3 gains the `check_register()` task. No finding ID renumbered; no design figure changed.
- **v1.6** — §3 records the render-gate near-miss closed in v1.4: the release gate had no pagination or footer check, and a fully broken twelve-document PDF set passed every structural check. No finding ID renumbered; no design figure changed.